Gebze Technical University

Department of Mechanical Engineering

ME 331 System Dynamics and Vibration

Damped Free Vibration

Single-degree-of-freedom systems

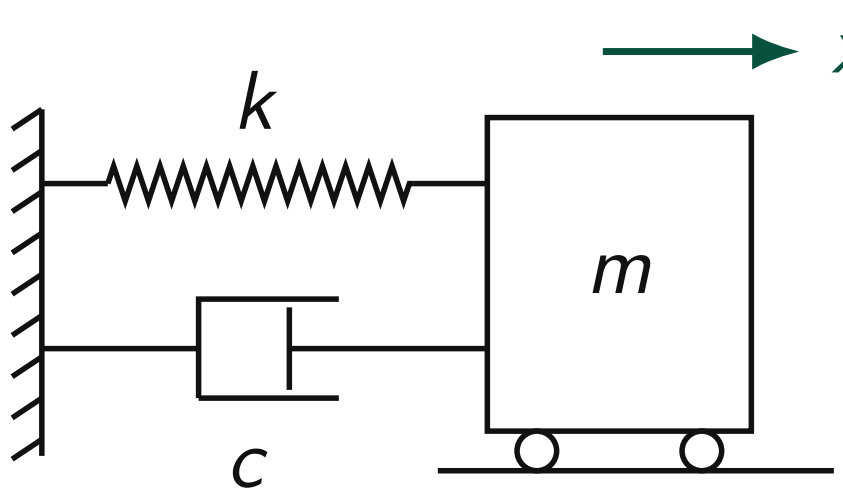
Saeed Lotfan

Today's topics

1. Damping models
2. Viscously damped free response
3. Measuring damping
4. Energy dissipation
5. Coulomb damping
6. Hysteretic damping
7. Comparison and checks

4.1 Damping models

Viscous force and equation of motion



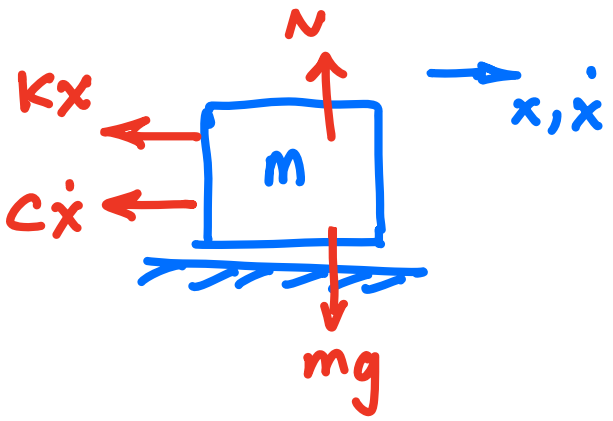
$$F_d = -c\dot{x}$$

c : $\mathrm{N\,s/m}$

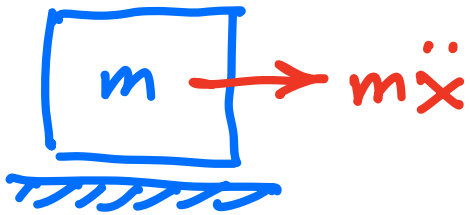
$$\boxed{m\ddot{x} + c\dot{x} + kx = 0}$$

Displacement from equilibrium. Damping opposes velocity.

Whiteboard notes



F.B.D



K.D

$$m\ddot{x} + kx = 0$$

$$\Sigma F \equiv \Sigma \mathcal{F}$$

$$-kx - c\dot{x} = m\ddot{x} \rightarrow m\ddot{x} + c\dot{x} + kx = 0$$

Whiteboard notes

$m\ddot{x} + c\dot{x} + Kx = 0$

$x(t) = Ce^{st} \rightarrow \dot{x}(t) = Cse^{st} = sx(t) \rightarrow \ddot{x}(t) = Cs^2e^{st} = s^2x(t)$

$ms^2Ce^{st} + csCe^{st} + KCe^{st} = 0$

$Ce^{st}(ms^2 + cs + k) = 0 \rightarrow s_{1,2} = \dfrac{-c \pm \sqrt{c^2 - 4km}}{2m}$

$c = 0 \rightarrow s_{1,2} = \pm\dfrac{\sqrt{-km}}{m} = \pm\sqrt{-\dfrac{k}{m}} = \pm i\sqrt{\dfrac{k}{m}} = \pm i\omega_n$ undamped

If a) $c^2 - 4km > 0$

b) $c^2 - 4km < 0$

c) $c^2 - 4km = 0$

$s_{1,2} = -\dfrac{c}{2m} \pm \sqrt{\left(\dfrac{c}{2m}\right)^2 - \dfrac{k}{m}}$

Whiteboard notes

$$x(t) = C_1 e^{s_1 t} + C_2 e^{s_2 t}$$

a) $s_{1,2} \in \mathbb{R} \rightarrow x(t) = C_1 e^{s_1 t} + C_2 e^{s_2 t}$

b) $s_{1,2} \in \mathbb{C} \rightarrow x(t) = C_1 e^{st} + C_2 e^{s_2 t}$

$$= A e^{\mathrm{Re}(s_1)t} \cdot \cos \mathrm{Im}(s_1)t + B e^{\mathrm{Re}(s_1)t} \sin \mathrm{Im}(s_1)t + \dots$$

c) $s_{1,2} = -\frac{c}{2m} = s$

$$x(t) = C_1 e^{st} + C_2 t e^{st}$$

4.1 Damping models

Critical damping and damping ratio

$c^2 - 4km = 0$

$c_c = 2\sqrt{km}$

$$\omega_n = \sqrt{\frac{k}{m}}, \qquad c_c = 2\sqrt{km}, \qquad \zeta = \frac{c}{c_c}$$

underdamped

$c < c_c \quad \zeta < 1$

$c = c_c \quad \zeta = 1$

$c > c_c \quad \zeta > 1$

overdamped

$$\ddot{x} + 2\zeta\omega_n\dot{x} + \omega_n^2 x = 0$$

Try $x = Ce^{st}$. What changes when ζ crosses 1?

$m\ddot{x} + c\dot{x} + kx = 0 \rightarrow \ddot{x} + \frac{c}{m}\dot{x} + \frac{k}{m}x = 0$ (ω_n^2)

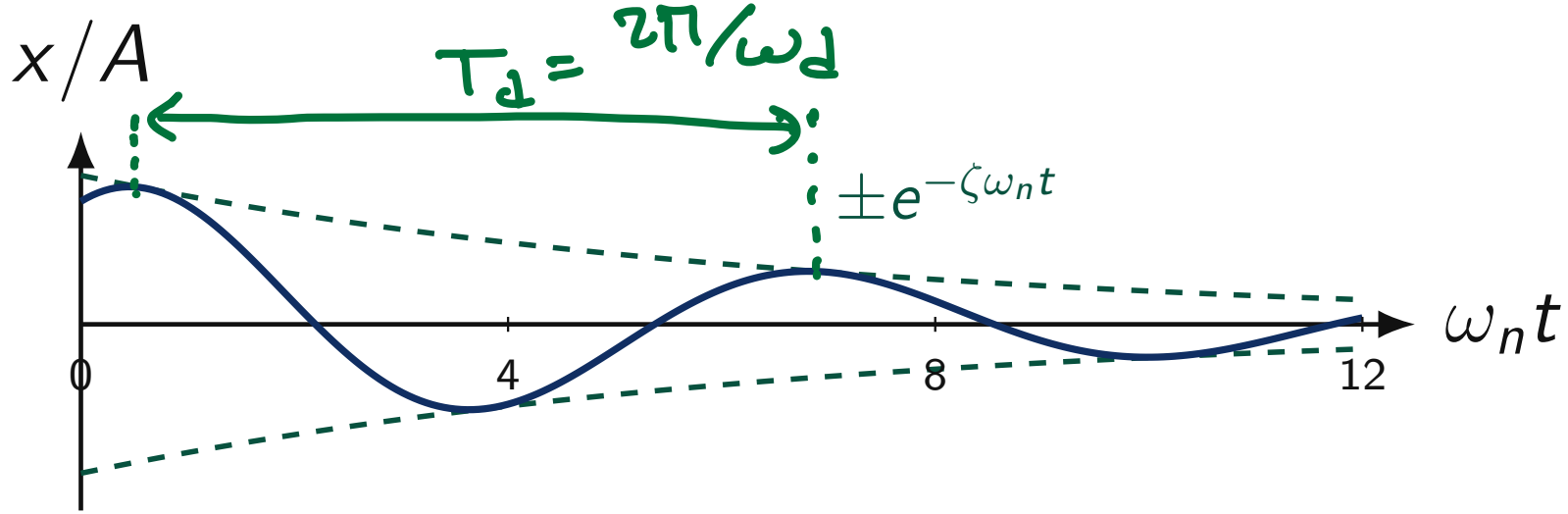
$\frac{c}{m} = \frac{\zeta c_c}{m} = \frac{\zeta 2\sqrt{km}}{m} = \zeta 2\sqrt{\frac{k}{m}} = 2\zeta\omega_n$

4.2 Viscously damped free response

Underdamped motion: frequency and envelope

undamped $x(t) = A\cos(\omega t - \phi)$

$$x = Ae^{-\zeta\omega_n t}\cos(\omega_d t - \phi), \qquad \omega_d = \omega_n\sqrt{1-\zeta^2}$$



$0 < \zeta < 1$ $\qquad T_d = 2\pi/\omega_d$ $\qquad$ Initial conditions set A, ϕ.

Whiteboard notes

$$\ddot{x} + 2\zeta\omega_n\dot{x} + \omega_n^2 x = 0 \qquad 0 < \zeta < 1,\ c < c_c\ ;\ c < 2\sqrt{km}$$

$$x = Ce^{st}$$

$$Ce^{st}\left(s^2 + 2\zeta\omega_n s + \omega_n^2\right) = 0 \qquad s_{1,2} = \frac{-\zeta\omega_n \pm \sqrt{4\zeta^2\omega_n^2 - 4\omega_n^2}}{2}$$

$$s_{1,2} = -\zeta\omega_n \pm \omega_n\sqrt{\zeta^2 - 1}$$

$$s_{1,2} = -\zeta\omega_n \pm i\omega_n\sqrt{1-\zeta^2} = \underbrace{-\zeta\omega_n}_{\mathrm{Re}(s)} \pm i\,\underbrace{\omega_n\sqrt{1-\zeta^2}}_{\mathrm{Im}(s)}$$

$$x(t) = C_1 e^{-\zeta\omega_n t} \cdot e^{i\omega_n\sqrt{1-\zeta^2}\,t} + C_2 e^{-\zeta\omega_n t} \cdot e^{-i\omega_n\sqrt{1-\zeta^2}\,t}$$

Whiteboard notes

$e^{i\alpha} = \cos\alpha + i\sin\alpha$

$$x(t) = e^{-\zeta\omega_n t}\left[C_1 e^{i\omega_n\sqrt{1-\zeta^2}\,t} + C_2 e^{-i\omega_n\sqrt{1-\zeta^2}\,t} \right]$$

$$= e^{-\zeta\omega_n t}\left[A\cos\omega_n\sqrt{1-\zeta^2}\,t + B\sin\omega_n\sqrt{1-\zeta^2}\,t \right]$$

2 Harmonics $\Longrightarrow$ summed into 1 (week 7)

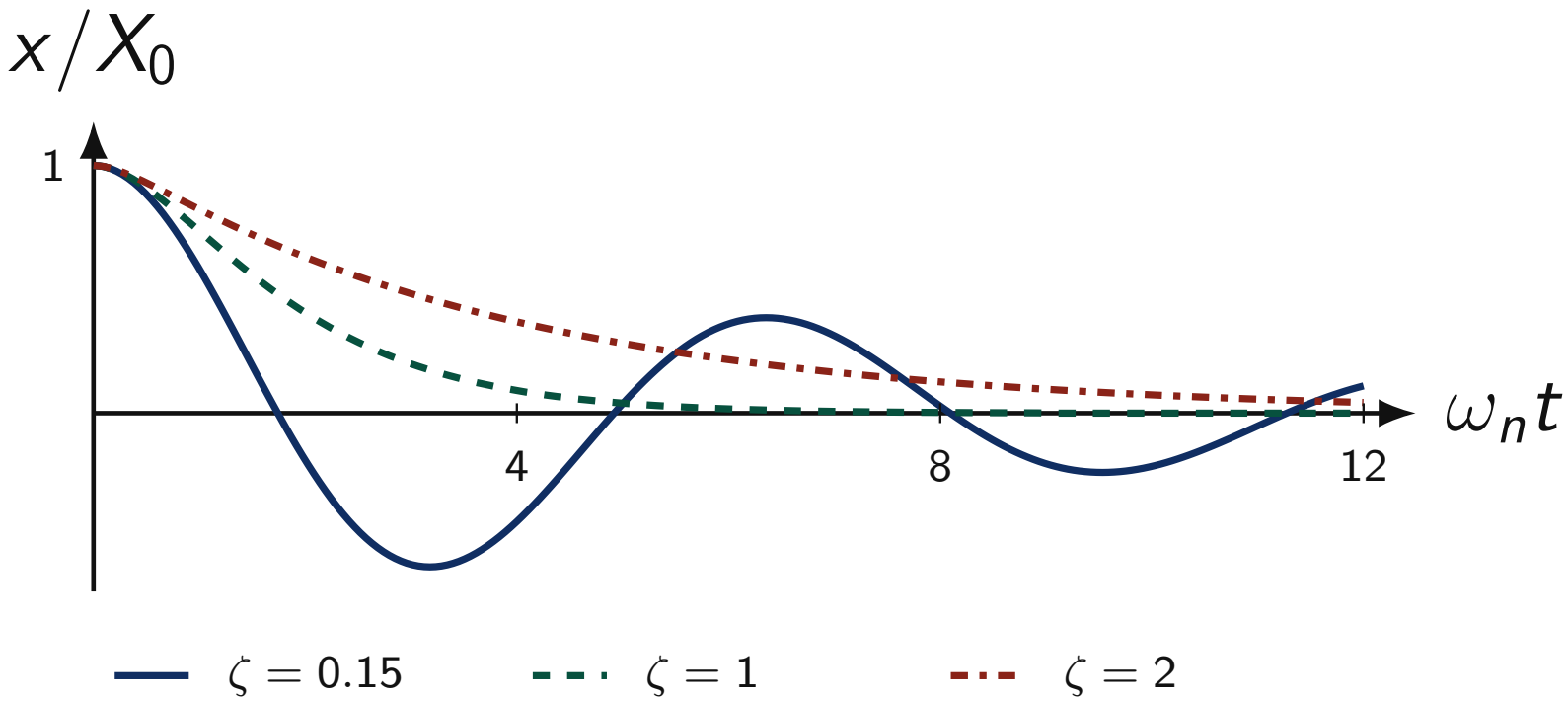
$$x(t) = X\underbrace{e^{-\zeta\omega_n t}}_{\text{decay part}}\cos\Big(\underbrace{\omega_n\sqrt{1-\zeta^2}}_{\omega_d}\,t - \phi\Big)$$

oscillation part

4.2 Viscously damped free response

Three damping cases, the same initial conditions

Fixed m, k $\quad x(0) = X_0 > 0 \quad \dot{x}(0) = 0$



Oscillatory Critical Overdamped

Whiteboard notes

$\zeta = 1 \qquad \ddot{x} + 2\omega_n \dot{x} + \omega_n^2 x = 0$

$s_{1,2} = \frac{-2\omega_n}{2} \pm \frac{\sqrt{4\omega_n^2 - 4\omega_n^2}}{2} = -\omega_n$

$x(t) = C_1 e^{-\omega_n t} + C_2 t e^{-\omega_n t}$

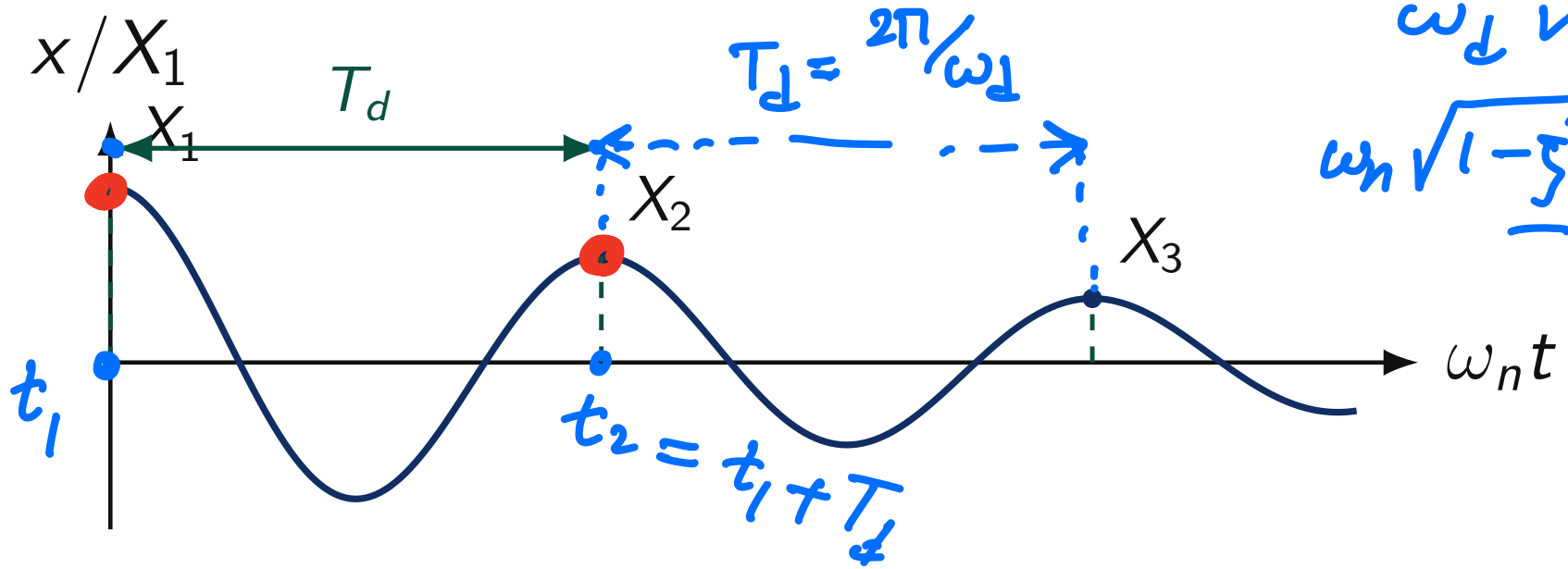
$\zeta > 1 \qquad s_{1,2} = -\zeta\omega_n \pm \frac{\sqrt{4\zeta^2\omega_n^2 - 4\omega_n^2}}{2} = -\zeta\omega_n \pm \omega_n\sqrt{\zeta^2 - 1}$

$x(t) = C_1 e^{(-\zeta\omega_n + \omega_n\sqrt{\zeta^2-1})t} + C_2 e^{(-\zeta\omega_n - \omega_n\sqrt{\zeta^2-1})t}$

$= \text{Real} + \text{Real}$

4.3 Measuring damping

Logarithmic decrement from same-sign peaks



$$\frac{2\pi\zeta n}{\sqrt{1-\zeta^2}} = \delta = \frac{1}{n}\ln\left(\frac{X_j}{X_{j+n}}\right), \qquad \zeta = \frac{\delta}{\sqrt{4\pi^2+\delta^2}}$$

Count complete cycles. Measure amplitudes from equilibrium.

Whiteboard notes

$$x(t) = A e^{-\zeta\omega_n t} \cos(\omega_d t - \phi)$$

$$x(t_1) = A e^{-\zeta\omega_n t_1} \cos(\omega_d t_1 - \phi) = X_1$$

$$x(t_2) = A e^{-\zeta\omega_n t_2} \underbrace{\cos(\omega_d t_2 - \phi)}_{\cos(\omega_d t_1 + \underbrace{\omega_d T_d}_{2\pi} - \phi)} = X_2$$

$$t_2 = t_1 + T_d$$

$$\cos(\omega_d t_1 + \omega_d T_d - \phi) = \cos(\omega_d t_1 - \phi)$$

$$\frac{X_1}{X_2} = \frac{\cancel{A} e^{-\zeta\omega_n t_1} \cancel{\cos(\omega_d t_1 - \phi)}}{\cancel{A} e^{-\zeta\omega_n t_2} \cancel{\cos(\omega_d t_1 - \phi)}} = e^{-\zeta\omega_n \underbrace{(t_1 - t_2)}_{-T_d}}$$

Whiteboard notes

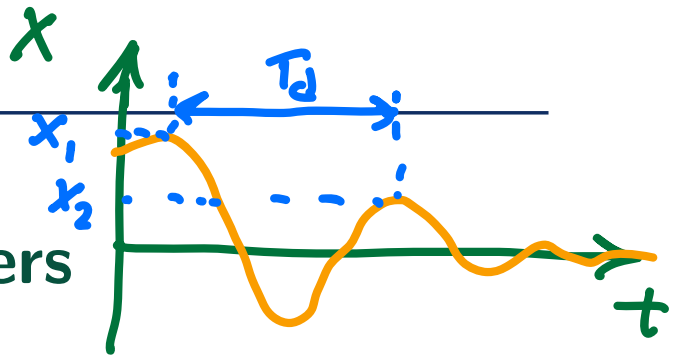
$$\frac{x_j}{x_{j+n}}\frac{x_1}{x_2} = e^{\zeta\omega_n(\tau_d n)} \rightarrow \ln\frac{x_j}{x_{j+n}}\frac{x_1}{x_2} = n\zeta\omega_n\frac{2\pi}{\omega_d}$$

$$= n\zeta\frac{\omega_n 2\pi}{\omega_n\sqrt{1-\zeta^2}}$$

Logarithmic Decrement

$$\ln\frac{x_j}{x_{j+n}}\frac{x_1}{x_2} = \frac{n2\pi\zeta}{\sqrt{1-\zeta^2}} \rightarrow \zeta$$

4.3 Measuring damping



From a decay record to model parameters

$$\omega_d = \frac{2\pi}{T_d}, \qquad \omega_n = \frac{\omega_d}{\sqrt{1-\zeta^2}}$$

$$\delta = \ln \frac{x_1}{x_2}$$

If the mass is known:

$$k = m\omega_n^2, \qquad c = 2\zeta m \omega_n$$

From the free-decay graph, which system parameters can we determine without knowing the mass?

4.4 Energy dissipation

$P_{loss} = F \cdot v = c\dot{x} \cdot \dot{x}$

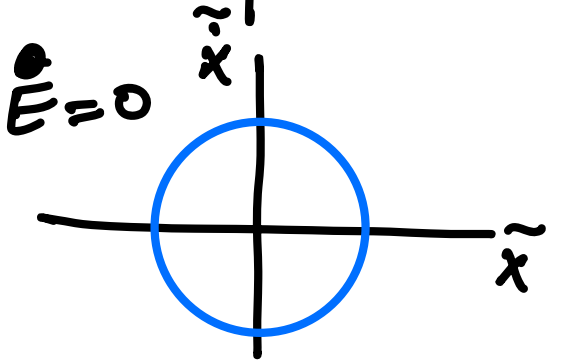
Mechanical energy decreases during free motion

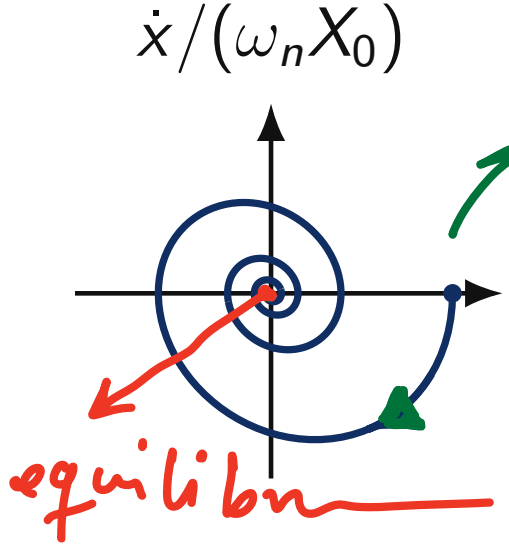
$$E = \frac{1}{2}m\dot{x}^2 + \frac{1}{2}kx^2$$

$$\boxed{\dot{E} = -c\dot{x}^2 \leq 0}$$

$c\dot{x}^2$

circle for undamped case





Initial conditions

$x(0) = X_0$

$\dot{x}(0) = 0$

equilibrium $x = \dot{x} = 0$

Dissipated power: $c\dot{x}^2$. When is it zero or greatest?

Whiteboard notes

$$m\ddot{x} + c\dot{x} + kx = 0$$

$$s_{1,2} = \frac{-c}{2m} \pm \sqrt{\frac{c^2 - 4km}{4m^2}}$$

If a) > 0 → overdamped
b) < 0 → underdamped
c) $= 0$ → critically dam—

$$c_c^2 - 4km \rightarrow \quad c_c = 2\sqrt{km}, \quad \zeta = c/c_c, \quad \omega_d = \omega_n\sqrt{1-\zeta^2}$$

damping ration

Whiteboard notes

$$\dot{E} = -c\dot{x}^2 \longrightarrow \Delta W = \int_0^{T_d} c\dot{x}^2 \, dt$$

assume $x(t) = X \cos \omega_d t$ → a harmonic motion

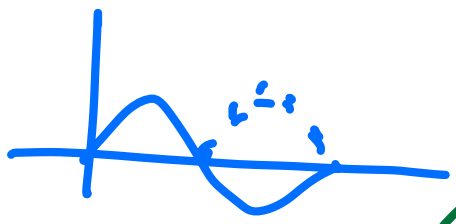
$$\Delta W = \int_0^{2\pi/\omega_d} c X^2 \omega_d^2 \sin^2 \omega_d t \, dt$$

$u = \omega_d t$

$du = \omega_d \, dt$

$$= \int_0^{2\pi} c X^2 \omega_d \sin^2 u \, du$$

$$= \int_0^{2\pi} \frac{1 - \cos 2u}{2} du = \pi$$

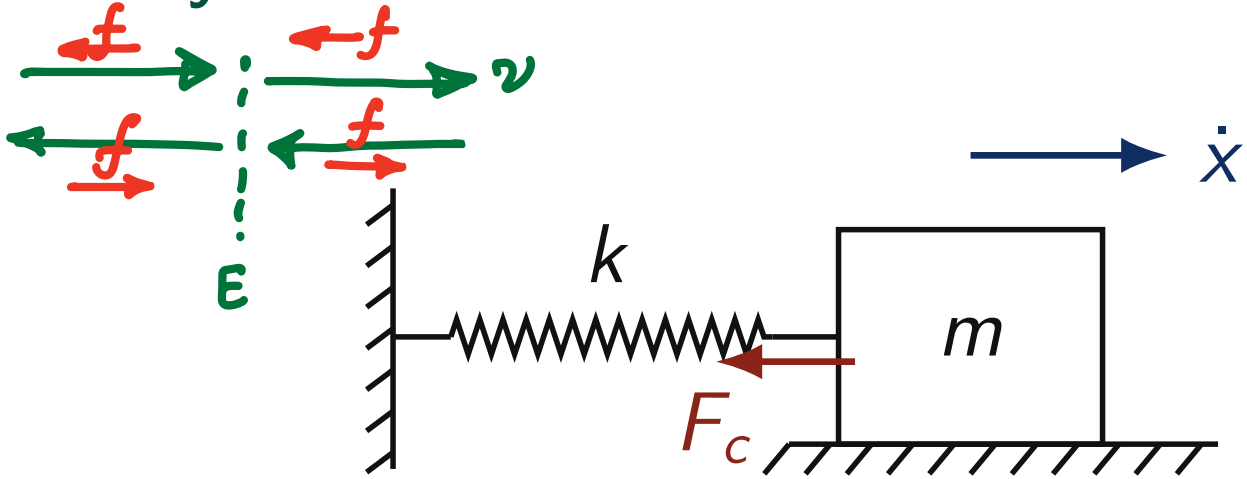


$\pi c \omega_d X^2$

Energy loss in one period due to $c\dot{x}$

4.5 Coulomb damping (dry friction)

Dry friction reverses with the direction of sliding



$$F_c = \mu_k N$$

metal-metal $\mu = 0.1 - 0.3$
rubber-metal $\mu \approx 1$

$$\begin{cases} m\ddot{x} + kx + F_c = 0, & \dot{x} > 0, \\ m\ddot{x} + kx - F_c = 0, & \dot{x} < 0. \end{cases}$$

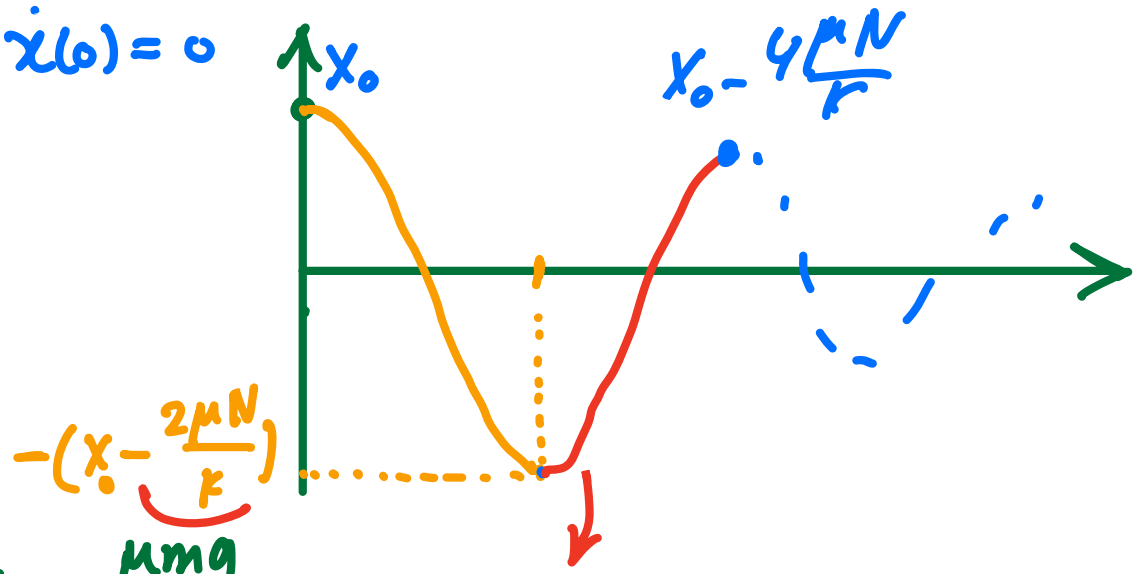
Signum(y) = $\begin{cases} 1 & y > 0 \\ -1 & y < 0 \end{cases}$

Solve one sliding interval, then check the turning point.

$$m\ddot{x} + kx + \mathrm{sgn}(\dot{x}) F_c = 0$$

Whiteboard notes

Solve for IC: $x(0) = X_0$ $\dot{x}(0) = 0$



$\dot{x} < 0$

$m\ddot{x} + Kx - \mu mg = 0$

$m\ddot{x} + Kx = \mu mg$

$x(t) = A_1 \sin \omega_n t + A_2 \cos \omega_n t + \frac{\mu mg}{K}$

$m\ddot{x} + Kx + \mu mg = 0$

$x(t) = B_1 \sin \omega_n t + B_2 \cos \omega_n t - \frac{\mu mg}{K}$

$x(0) = X_0 \rightarrow \quad A_2 + \frac{\mu mg}{K} = X_0$

$\dot{x}(0) = 0 \rightarrow \quad A_1 = 0$

$$x(t) = \left(X_0 - \frac{\mu mg}{K}\right) \cos \omega_n t + \frac{\mu mg}{K} \qquad 0 < t < \frac{T}{2}$$

Whiteboard notes

$$x\left(\frac{T}{2}\right) = B_1 \sin\frac{2\pi}{T}\cdot\frac{T}{2} + B_2 \underbrace{\cos\pi}_{-1} - \frac{\mu mg}{\kappa} = -\left(x_0 - \frac{2\mu N}{k}\right)$$

$\dot{x} \Rightarrow \quad B_1 = 0$

$$B_2 = +\left(x_0 - \frac{2\mu N}{k}\right) - \frac{\mu mg}{\kappa}$$

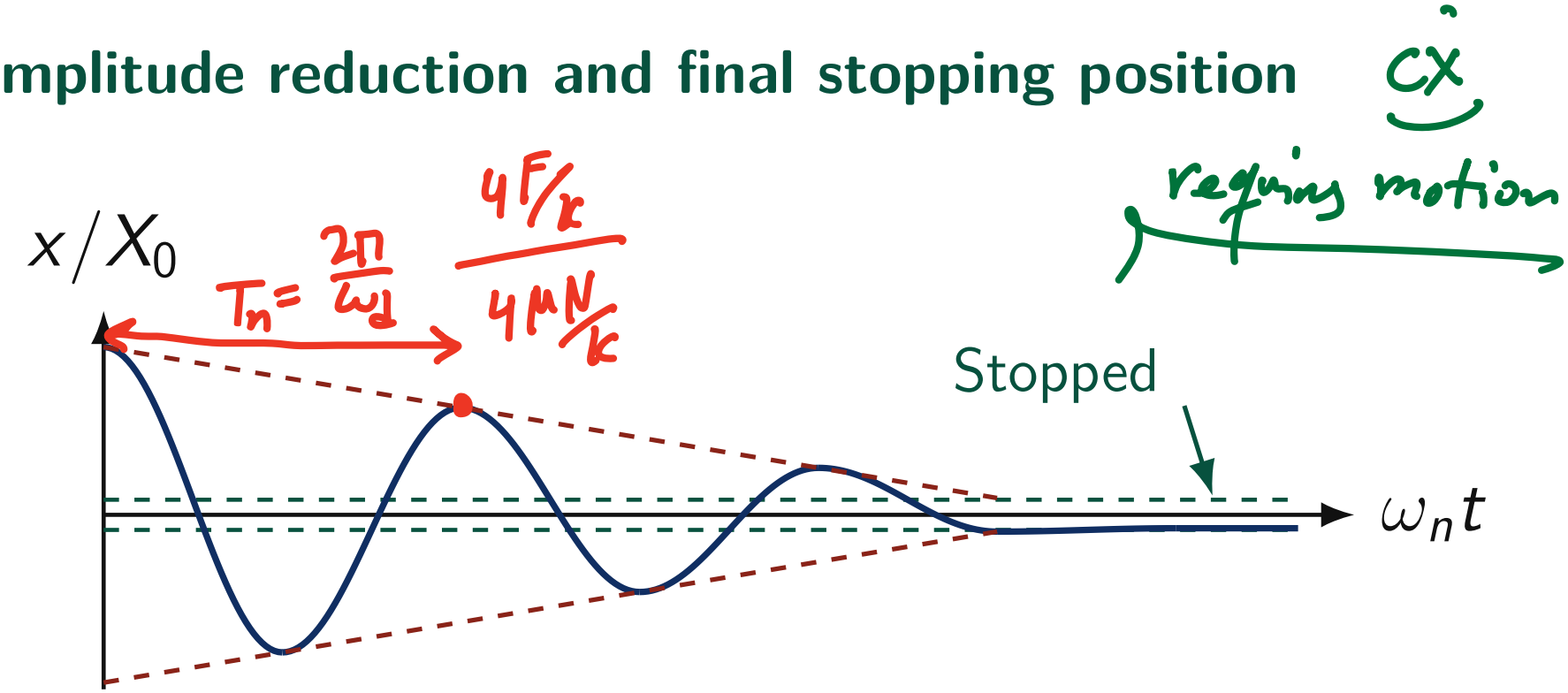
$$B_2 = x_0 - \frac{3\mu N}{\kappa}$$

$$x(t) = \left(x_0 - \frac{3\mu N}{\kappa}\right)\cos\omega_n t - \frac{\mu N}{\kappa}$$

$$x(T) = x_0 - \frac{4\mu N}{\kappa}$$

4.5 Coulomb damping

Amplitude reduction and final stopping position



$$\Delta X_{\text{half-cycle}} = \frac{2F_c}{k}, \qquad \Delta X_{\text{cycle}} = \frac{4F_c}{k}$$

At rest: $|kx_{\text{stop}}| \leq \mu_s N$. The final position need not be zero.

4.5 Coulomb damping

Viscous and Coulomb damping: comparison

Feature	Viscous	Coulomb
Force	$-c\dot{x}$	$-F_c \operatorname{sgn}(\dot{x})$
Equation	Linear	Nonlinear overall
Oscillation	$\omega_d = \omega_n\sqrt{1-\zeta^2}$ for $\zeta < 1$	$\omega_n = \sqrt{k/m}$ while sliding
Peak decay	Exponential decay	$4F_c/k$ decrease per cycle
Stopping	Asymptotic return	Finite-time sticking
Final position	Static equilibrium	May remain at $x \neq 0$

Whiteboard notes

ICs: $x(0) = X_0 \qquad \dot{x}(0) = 0$

$$\Delta X_{cycle} = \frac{4\mu N}{k}$$

$$k x_{stop} \leq \mu N$$

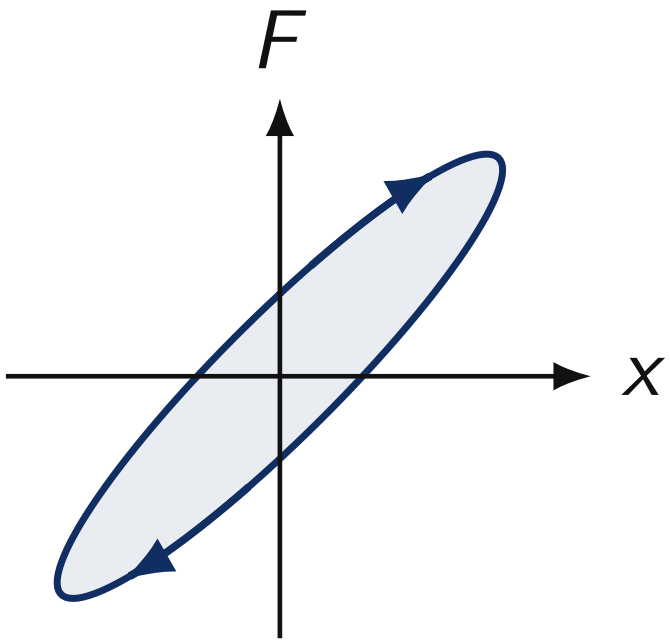
$$x_0 - n\left(\frac{2\mu N}{k}\right) \leq \frac{\mu N}{k}$$

$$n \geq \frac{x_0 - \frac{\mu N}{k}}{\frac{2\mu N}{k}}$$

n number of half cycles that the mass vibrates before stop due to x_0

4.6 Hysteretic damping

A force–displacement loop measures energy loss



Area $= \Delta W$

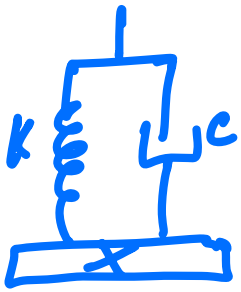
$$\Delta W = \oint F\, dx$$

$$\beta = \frac{\Delta W}{\pi k X^2}$$

$$\zeta_{\text{eq}} \approx \frac{\beta}{2}$$

Equivalent viscous damping assumes nearly harmonic motion and small loss.

Whiteboard notes



$F = Kx(t) + C\dot{x}(t)$ $\quad x(t) = X\sin\omega t$

$F(t) = KX\sin\omega t + CX\omega\cos\omega t$

$= KX\sin\omega t \pm CX\omega\sqrt{1 - \sin^2\omega t}$

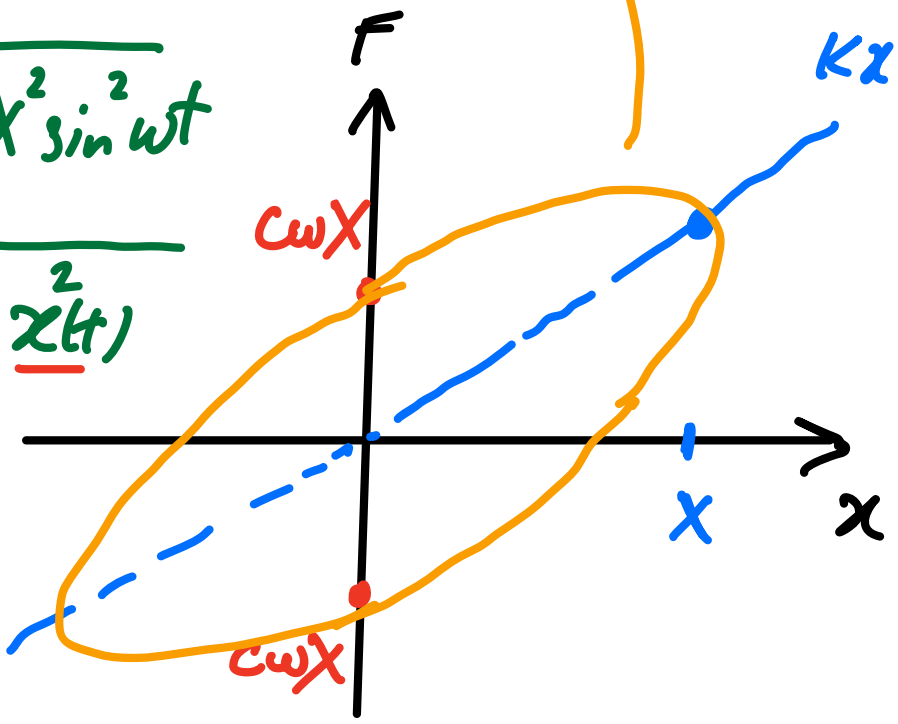
$= Kx(t) \pm C\omega\sqrt{X^2 - X^2\sin^2\omega t}$

$F = Kx(t) \pm C\omega\sqrt{X^2 - x^2(t)}$

$x(t)_{max} = X$

$F = KX \pm C\omega\sqrt{X^2 - X^2}$

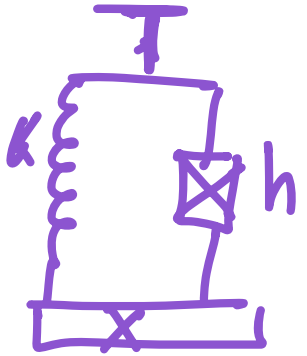
Energy loss in $C\dot{x}$ is elliptic



Whiteboard notes

$\Delta W = \pi c \omega X^2$ $\quad c\omega = h$ definition of Hysteretic damping

$\Delta W = \pi h X^2$



$f(t) = Kx(t) + ?$ $\quad f(t) = Kx(t) + ihx(t)$

$f(t) = Kx(t) + c\dot{x}(t)$; $x(t) = X \sin \omega t$; $x(t) = X e^{i\omega t}$

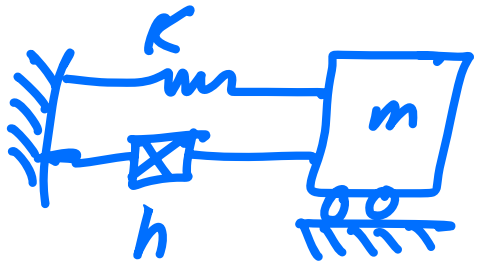
$= KX e^{i\omega t} + cX e^{i\omega t} \cdot i\omega = \left(KX + \underbrace{ic\omega}_{h} X\right) e^{i\omega t}$

$c\dot{x} \equiv ihx$

Whiteboard notes

$$\ddot{x} + 2\zeta\omega_n\dot{x} + \omega_n^2 x = 0$$

$$m\ddot{x} + ih\dot{x} + kx = 0$$



$$\zeta = \frac{\beta}{2}$$

$$\beta = h/k$$

$$x(t) = Ce^{st}$$

$$(ms^2 + ih + k)Ce^{st} = 0$$

$$s^2 = \pm\sqrt{\frac{ih+k}{m}}$$

$$c\dot{x} \rightarrow \pi X^2\omega c = \Delta W \rightarrow \delta = \frac{2\pi\zeta}{\sqrt{1-\zeta^2}}$$

$$ihx \rightarrow \pi X^2 h = \Delta W$$

$$\pi X^2 h = \Delta W \qquad \frac{1}{2}kX_j^2 - \frac{\pi h X_j^2}{4} - \frac{\pi h X_{j+1/2}^2}{4} = \frac{kX_{j+1/2}^2}{2}$$

$$\hookrightarrow X_j/X_{j+1/2} = \sqrt{\frac{2+\pi h/k}{2-\pi h/k}}$$

Whiteboard notes

$\frac{h}{k} = \beta \rightarrow$ hysteritic effect.

$$\frac{x_j}{x_{j+\frac{1}{2}}} = \sqrt{\frac{2+\pi\beta}{2-\pi\beta}} \rightarrow \ln\sqrt{\frac{2+\pi\beta}{2-\pi\beta}}$$

$\zeta \ll 1 \rightarrow c\dot{x} \rightarrow \delta \approx 2\pi\zeta$

$ih\dot{x} \rightarrow \delta \approx \pi\beta$

If reduction in half cycle of both is close to each oth. $\rightarrow \zeta \approx \frac{\beta}{2}$

4.7 Comparison and checks

Three mechanisms and their signatures

Viscous

Force proportional to velocity; exponential free-response envelope.

Coulomb

Constant sliding-friction magnitude; equal peak decrements; finite stopping.

Hysteretic

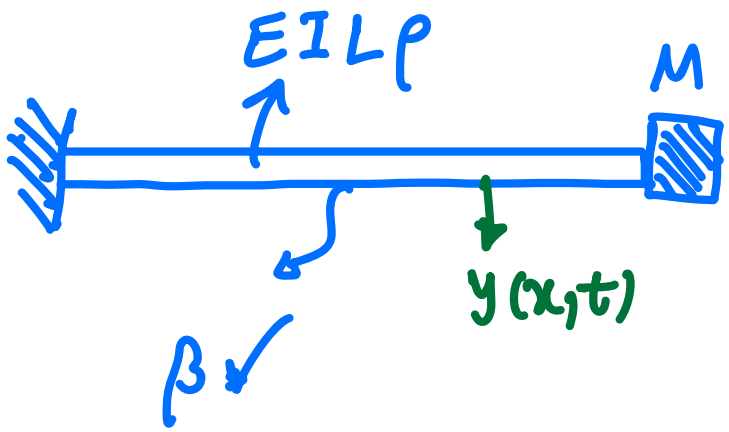
Material loop loss; approximately constant fractional energy loss per cycle.

4.7 Comparison and checks

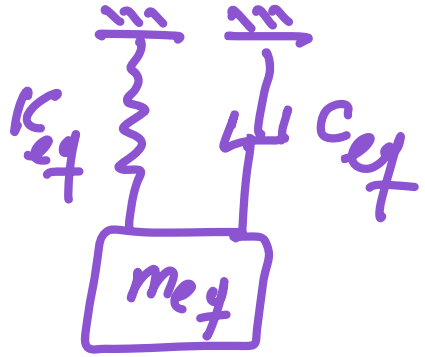
Before accepting a response

- Does the response satisfy x_0 and v_0?
- Are ω_n, ω_d, c, and ζ distinguished?
- Does the predicted energy behavior fit the model?
- Is the stopping or equivalence assumption justified?

Whiteboard notes







$y(x,t) = \psi(x) q(t)$ $\quad \psi(x) = \dfrac{x^2(3L-x)}{2L^3}$ → when BCs change you have to change this function!

$$T = \frac{1}{2} M \dot{q}^2 + \frac{1}{2} \int_0^L \rho A \psi^2 \dot{q}^2 \, dx$$

↳ m_{eq} ✓ $\quad M + \frac{33}{140} m_b$

Whiteboard notes

$$U = \frac{1}{2}\int_0^L EI\,\psi''^2 q^2\,dx$$

please do the integral.

$$k_{eq} = \frac{3EI}{L^3}$$

$$m_{eq}\ddot{q} + c_{eq}\dot{q} + k_{eq}q = 0$$

$$\beta = \frac{h}{k_{eq}}$$

$$\zeta = \frac{c_{eq}}{2\sqrt{k_{eq} m_{eq}}}$$

c_{eq}

$$\beta \rightarrow \zeta = \frac{\beta}{2}$$

Whiteboard notes

Whiteboard notes

Whiteboard notes